

# INDIAN STATISTICAL INSTITUTE

Project Linked Person — Selection 2026–2027

Practical Examination

Total: 14 + 16 + 10 = 40 marks      Duration: 90 minutes

## EXAMINATION INSTRUCTIONS

1. You have root on a virtual machine. It is connected to no real network. Log in with `ssh candidate@192.168.56.10`.
2. Each question has four parts: *the problem*, *how to reproduce it*, *what to fix*, and *how to check your work* (so that you do not have to guess whether you are finished).
3. Partial credit would be given if you can explain the problem and the reason behind, even if you cannot fix it.

### Q1. Disk full on the lab file server.

[14 marks]

*Reported by Debarshi, research scholar:* “I cannot copy anything into `/srv/data`, it says the disk is full, and nothing I can see in there accounts for that much space. Submitting a job into `/srv/spool` *also* says the disk is full, but `df -h` shows that one has plenty of space left. That part makes no sense to me.”

**Background:** this server has three separate disks. `/srv/data` and `/srv/spool` look like ordinary folders, but each one is really its own small disk, attached at that path (a *mount point*). The rest of the system lives on a third, much larger disk.

|                         |                         |             |
|-------------------------|-------------------------|-------------|
| <code>/</code>          | the system disk         | about 24 GB |
| <code>/srv/data</code>  | the project data disk   | 1 GB        |
| <code>/srv/spool</code> | the job-submission disk | 48 MB       |

Though the two complaints look alike, they are **two unrelated faults on two different disks**, with different causes. Fixing one does nothing for the other.

#### (a) `/srv/data` is full

[11 marks]

`/srv/data` is at 100%, and the files you can see there do not add up to the space reported.

*Reproduce:*

|                                    |                                       |
|------------------------------------|---------------------------------------|
| <code>df -h /srv/data</code>       | # 100% used                           |
| <code>sudo du -sh /srv/data</code> | # much smaller than df says is in use |

*Fix:* bring `/srv/data` back to **under 10% used** (the `Use%` column of `df -h`). There is more than one reason this disk is full: freeing the largest files may not work. Observe the differences shown in the output of `df` and `du`. Explain the reason behind the difference.

*Check:*

|                                   |                                            |
|-----------------------------------|--------------------------------------------|
| <code>df -h /srv/data</code>      | # Use% under 10%                           |
| <code>ls /srv/data/current</code> | # five survey-block CSV files, still there |

(b) /srv/spool refuses new files

[3 marks]

New files cannot be created on /srv/spool, *although df -h reports free space on it*. Something other than free space has run out. This has nothing to do with /srv/data.

*Reproduce:*

```
df -h /srv/spool          # shows space free
touch /srv/spool/incoming/testjob  # "No space left on device"
```

*Fix:* make /srv/spool able to accept new files again. The current problem is not due to a lack of free space in the disk. Explain why /srv/spool reported “No space left on device” while df -h showed free space.

*Check:*

```
touch /srv/spool/incoming/t && echo OK && rm /srv/spool/incoming/t
```

Constraints

- Nothing under /srv/data/current may be deleted or changed.
- /srv/data/archive/README.txt says what is safe to remove there. Read it before deleting anything.

Q2. Shared project folder unusable.

[16 marks]

*Reported for the ACMU Lab group:* “Buddhadev cannot save anything into shared/ which he could last month. And Chandrima from the internal audit cell needs to *read* that folder for the annual review, and cannot see anything at all.”

**Background: who is who.** /srv/projects/acmu/ (and shared/ inside it) belongs to the acmu group. Anyone who is neither a member nor the auditor should have no access at all.

|           |                              |                                |
|-----------|------------------------------|--------------------------------|
| arnab     | member of acmu               | read and write everything      |
| buddhadev | member of acmu               | read and write everything      |
| chandrima | auditor, <b>not</b> a member | read everything, write nothing |

To see the current state:

```
cd /srv/projects/acmu
stat -c '%a %U:%G %n' . shared shared/notes.md
id buddhadev
getfacl /srv/projects/acmu/shared
```

Underneath the two complaints are **three independent faults**, each with its own cause and fix, each marked on its own. Take them in any order. **Making the fixes will need root (sudo)** — changing group membership, permissions and access rules all do. **But test as the users named, not as root:** root can do everything, so a check run as root proves nothing.

(a) Buddhadev is no longer in the acmu group

[5 marks]

*Reproduce:* Check id buddhadev where acmu is missing. *Fix:* put him back, without disturbing his other groups. Keep checking with id buddhadev until it shows acmu again.

**(b) The shared folder does not let the group write** [8 marks]

Members of `acmu` cannot create files in the folder at all. Arnab is still a member, so test as `arnab`:

```
sudo -u arnab touch /srv/projects/acmu/shared/t1      # Permission denied
stat -c '%a %U:%G %n' /srv/projects/acmu /srv/projects/acmu/shared
```

*Fix:* both directories must let the `acmu` group read, write and enter them, and give everyone else no write access.

```
sudo -u arnab touch /srv/projects/acmu/shared/t1 && echo "write OK"
sudo rm -f /srv/projects/acmu/shared/t1
```

**(c) The auditor cannot read the folder** [3 marks]

*Fix:* grant read-only access to everything under `/srv/projects/acmu/`, including files created there in future, and do it *without* adding `chandrima` to the group.

```
sudo -u chandrima cat /srv/projects/acmu/shared/notes.md # must work
sudo -u chandrima touch /srv/projects/acmu/shared/t2      # must FAIL
```

**When all three are done**

Buddhadev's complaint needs (a) and (b) together:

```
sudo -u buddhadev touch /srv/projects/acmu/shared/t3      # must work
sudo -u buddhadev cat /srv/projects/acmu/shared/notes.md  # must work
```

**Constraints**

- **Chandrima must not be put into the `acmu` group.** Audit policy forbids putting auditors into the groups they audit: read everything, write nothing.
- No `chmod 777`, anywhere. This is a multi-user machine.

**Q3. Cannot log in to the ACMU lab server.** [10 marks]

*Reported by Debarshi, research scholar:* "I am supposed to log in to the lab server as `acmusrv`. It worked last week. Now it just says *Permission denied*, and then asks me for a password I was never given. I have not touched anything since."

`ssh acmulab` should log you in to the account `acmusrv` on the lab server using the key `~/.ssh/id_acmu`, **without a password**. Two things are wrong, and they produce the **same message**: `Permission denied (publickey)`. Identify both of them.

**How to reproduce it**

```
ssh acmulab
cat ~/.ssh/config
ls -l ~/.ssh/
```

**What to fix** [10 marks]

**3.1 Make the `acmulab` alias use the right key.** The `~/.ssh/config` file is corrupted. Fix it so that `ssh acmulab` succeeds without a password prompt.

**3.2 Make the key usable.** Find out why `ssh` still refuses to use the key, and fix it.

## How to check you have fixed it

```
ssh acmulab whoami          # prints: acmusrv
ls -l ~/.ssh/id_acmu       # -rw----- (0600)
```

If that prints `acmusrv` without asking you for anything, the key is doing the work.